Bihar Engineering University, Patna

B.Tech 8th Semester Examination, 2025

Course: B.Tech
Code: 100817

Subject: Cloud Computing

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only):- **[2 x 7 = 14]**

(a) Which of the following is the basis for cloud computing?
(i) Modulation. (ii) Hybridization.
(iii) Virtualization. (iv) Transmutation.

(b) PaaS stands for?
(i) Platform as a service. (ii) Platform as a software.
(iii) Parallel as a service. (iv) None of the given options.

(c) Which of the following is examples of SaaS cloud computing?
(i) Dropbox. (ii) Salesforce.
(iii) Google workshop. (iv) All of the given options.

(d) What is the primary purpose of a hypervisor in virtualization?
(i) To create virtual hard disks.
(ii) To manage network connections.
(iii) To allocate hardware resources and manage virtual machines (VM).
(iv) To manage the guest operating systems (OS).

(e) What is the most important concern of cloud computing?
(i) Cost. (ii) Security.
(iii) Space. (iv) Platform.

(f) Which of the following is the most appropriate basis for utility computing?
(i) Virtual isolation model. (ii) Grid computing model.
(iii) Pay-per-use model. (iv) Service-oriented architecture.

(g) Which of the following is the correct relationship of cloud services to customers?
(i) One-one. (ii) One-many.
(iii) Many-one. (iv) Many-many.

(h) Which of the following is the most important use of cloud computing?
(i) Data storage. (ii) Data privacy.
(iii) Reliability. (iv) All of the given options.

(i) What is a common security concern related to multi-tenancy in cloud computing?
(i) Lack of internet connectivity. (ii) Insufficient processing power.
(iii) Data segregation. (iv) Limited scalability.

(j) What is the name of a security attack, where an attacker intercepts and alters communication between two parties in a cloud environment?
(i) Brute force attack. (ii) Man-in-the-Middle (MitM) attack.
(iii) Drive-by attacks. (iv) Eavesdropping attack.

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Q.2 (a) Describe the four deployment models of cloud computing. [8]

(b) Write the fundamental differences between Microsoft Azure storage service and Amazon S3 service. [6]

Q.3 (a) Describe with diagrams the roles of hypervisors in traditional, hybrid, and hosted VM architectures. [9]

(b) What are the different levels of tenant isolation required in a multi-tenant cloud architecture? [5]

Q.4 (a) Consider the following terms in service level agreement (SLA) between a certain email service provider and its user. [8]

- Response SLA time of service requests: must acknowledge and respond to any 'request from user' within 1 hour in 24×7 duration.
- Resolution SLA time of critical incidents (like email server outage): must resolve any 'incident reported by user' within 4 hours in 24×7 duration; 'non-critical issues' such as email set up must be resolved within 24 hours.
- Uptime guarantee: must operate 99.9% of time each month.

In a certain unfortunate scenario, one user sends an alert at 10:00AM reporting that email server is down. The support team acknowledges that 'incident' by 11:00AM, works on the problem and resolves it by 2:00PM.

Explain whether the service level agreement is fulfilled or not.

(b) (i) What is elasticity in service-oriented architecture? [6]

(ii) With diagram, explain how elasticity is rapidly achieved by cloud services.

Q.5 (a) Describe spot instance technique in provisioning of cloud resources. [8]

(b) (i) What is over-provisioning of cloud resources? [6]

(ii) Explain whether scalability of cloud services is achieved or not by over-provisioning.

Q.6 (a) Give a performance comparison of first come first service (FCFS), short job first schedule (SJF), and MAX-MIN task scheduling algorithm in a cloud environment; consider the following performance metrics for this question: total execution time (TET), total waiting time (TWT), and total finish time (TFT). [9]

(b) (i) What is load balancing? [5]

(ii) How does load balancing work in a cloud environment?

Q.7 (a) In the era when smartphones are becoming more and more resourceful, is it still justified to offload users' computation tasks in mobile cloud computing? Give reasons. [5]

(b) (i) What is containerization? [9]

(ii) How containers differ from VM?

(iii) What is the role of containerization in edge computing?

Q.8 (a) Write down the impact of the following attacks on the cloud services. [9]

(i) Distributed denial of service (DDoS) attack;

(ii) SQL injection attack;

(iii) Cross-site scripting (XSS) attack.

(b) How VM security is compromised by VM side-channel attack in the traditional hypervisor model? [5]

Q.9 Write short notes on *any two* of the following:- [7 x 2 = 14]

(a) Cloud service models.

(b) Virtual machine monitor (VMM).

(c) Security-as-a-Service (SecaaS) model.

(d) Fog computing.

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